

GHGs (greenhouse gases) emission and economic analysis of a GCRES (grid-connected renewable energy system) in the arid region, Algeria



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ABSTRACT

This paper presents a method for economic evaluation and GHGs (greenhouse gases) emissions calculation from a GCRES (grid-connected renewable energy system). An investigation is made on large-scale operations of 67 MWh/day GCRES. A comparison is performed between a GCRES and a standard grid operation focusing on environmental and economic impacts. Emissions and the Renewable energy generation fraction (RF) of total energy consumption are calculated as the main environmental indicators. Costs including NPC (net present cost), COE (cost of energy) and payback period are calculated as the economic indicators. Using the hourly mean global solar irradiance, temperature and wind speed data relative to In Salah and Adrar locations characterized by an arid and hot climate according to the Koppen–Geiger climate classification, a long-term continuous implementation of hybrid renewable energy systems are simulated using HOMER software and are discussed. As results, it is observed that a GCRES reduce 30% and 35% of GHGs emission, and 81% and 76% of COE during the operation phase respectively for In Salah and Adrar.

Investments in GCRES should be considered only by planning to produce parts of the equipment locally, which leads to significantly reduce the costs and, consequently, the emissions.

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1. Introduction

World energy consumption has been growing by about 2.3% per year [1]. On the other hand, the use of fossil fuels raises serious environmental concerns. The burning of fossil fuels produces around 21.3 billion tons (21.3 Gigatons) of carbon dioxide (CO₂) per year, but it is estimated that natural processes can only absorb about half of that amount, so there is a net increase of 10.65 billion tons of atmospheric carbon dioxide per year (one ton of atmospheric carbon dioxide is equivalent to 44/12 or 3.7 ton of carbon dioxide) [2]. Carbon dioxide is one of the greenhouse gases that enhances radiative forcing and contributes to global warming, causing the average Earth surface temperature to rise in response, which in the opinion of the majority of climate scientists will cause major adverse effects. A global move towards the generation of renewable energy is therefore under way to help reduce global greenhouse gas emissions. So, the application of the hybrid renewable energy system to generate electrical power has certain

advantages: enables continuous supply of electrical power, it is free and available, does not pollute the environment (no emission of greenhouse gases), and counts among the renewable energy sources. It also contributes to sustainable development and growing pressure on the world's nations to reduce greenhouse gas emissions.

In this context, recently published shows the importance of the hybrid renewable energy system [3–11]. Mohammad Nezami et al. [3] have modeled a complete hybrid system including a photovoltaic array, a wind turbine, and storage batteries to determine the best approach for sizing the system to meet the electrical energy needs of a residential building in Tehran. The obtained results showed that the electrical cost of the hybrid system in Tehran is 0.62 US\$/kWh, which is 78% less expensive than a wind turbine system and 34% less expensive than a photovoltaic system.

Cao, Sunliang, and Kai Sirén [4] have studied matching indices taking the dynamic hybrid electrical and thermal grids information into account for the decision-making of nZEB (Net Zero Energy Building) on-site renewable energy systems. The obtained results showed that the developed methodology could seek an optimized balance between the objectives of maximizing the matching capability and minimizing the environmental/economic load. A

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Nomenclature

$C_{ann,tot}$	total annualized cost [\$ /yr]
CRF	capital recovery factor
E_{pv}	the energy generation by photovoltaic (Wh)
E_{ToT}	the total energy generation (Wh)
E_{WG}	the energy generation by wind generator (Wh)
H	hub height of the wind turbine generator (m)
I	interest rate [%];
R_{proj}	project lifetime [yrs].
v	wind speed (m/s)
z_0	surface roughness length (m)
z	anemometer height (m)

Subscripts

AC	alternating current
----	---------------------

Bwh	Köppen Classification System for hot and Dry Arid (Desert) climate
CO_2	carbon dioxide
COE	cost of energy
DC	continuing current
FIT	feed in tariff (%)
G	solar irradiation ($kWh\ m^2/day$)
GCRES	grid-connected renewable energy system
GHGs	greenhouse gases
NOx	Nitrogen oxide
NPC	net present cost (\$/year)
P	grid electricity price (\$/kWh)
PV	photovoltaic
RF	renewable energy generation fraction (%)
S	sellback rate (\$/kwh)
SO_2	Sulfur dioxide

PSO (particle swarm optimization) based on the PI controllers of a grid-connected inverter power supplied from a hybrid renewable energy system was studied by Ref. [5]. The system is composed of two renewable energy sources (wind turbine and photovoltaic – PV – solar panels) and two energy storage systems (battery and hydrogen system, integrated by fuel cell and electrolyzer). The obtained results showed that the online PSO-based PI controllers that optimize the algorithm based on the integral time absolute error index achieves the best response.

In Ref. [6] an advanced energy management strategy for a stand-alone hybrid energy system including a photovoltaic panel, a fuel cell, an electrolyzer, a battery bank and a super capacitor was considered. The simulation results confirmed the efficiency of the proposed control strategy, as it increases the reliability of the system and improves its energy balance.

In Ref. [7] research, an optimum design of the combined solar collector and geothermal heat pump system was studied. Their main purpose is analyzing the system from economical and technical points of view simultaneously. An optimization of two parabolic trough solar thermal power plants integrated with TES (thermal energy storage) and FBS (fuel backup system) has been performed by Boukelia, T. E., et al. [8].

The work of Stojiljkovic, Miro M et al. [9] consists of the greenhouse gases emission assessment in residential sector through buildings simulations and operation optimization studies.

Johansson, Daniella, et al. [10] have realized a comparative study of Fischer–Tropsch production and post-combustion CO_2 capture at an oil refinery. The obtained results showed that a low charge for CO_2 economically favors FT fuel production and the CO_2 capture confirms the greatest reduction in GHG emissions.

The paper of Baghdadi et al. [11] has investigated the performance of hybrid PV–Wind–Diesel–Battery configuration based on hourly measurements related to Adrar climate (southern Algeria). Data measurements of global solar radiation, ambient dry air temperature and wind speed over a period of one year have been used.

Note that Algeria is amongst the top ten countries in the world in terms of natural gas and oil production. Indeed, Algeria relies heavily on hydrocarbons, with 94% of energy currently coming from natural gas, representing 50% of the national GDP. Recent studies suggest that about 5% of the country's electricity comes from small hydropower plants while only 0.5%–1% comes from wind and solar energy [12] and [13].

Studies estimate that electrical energy use in Algeria will rise to 83 Terawatt-hour (TWh, $1\ TWh = 10^{12}$ Watts. hour) by 2020 and up to 150 TWh by 2030 [12] and [14]. Conversely, current projections estimate that the country's oil reserves will only cover the next 50 years while those of natural gas will only be available over the next 70 years [12] and [13]. Algeria thus faces a mounting challenge between its dependence on fossil fuels and its capacity for exploiting vast renewable sources. Some ambitious plans to develop renewable energy over the period of 2011–2030 have been put forward aspiring to generate 40% of local electricity by 2030 from solar and wind energy [3,14], with an estimated capacity of 22,000 Megawatts (MW) dedicating 12,000 MW to domestic use and 10,000 MW to exports at a cost of \$60 US billion.

A recent report showed that energy usage in Algeria is split between three sectors, industrial (24%), transport (33%), and residential and services (43%) [15]. The renewable energy mix of solar and wind energy as well as effective engineering applications can potentially contribute towards energy provision for these sectors and help move the country towards a more sustainable position in terms of energy provision and consumption [12].

Algeria is very well placed to be a major actor in the lucrative market of renewable energy. However, transition to more renewable energy use will need to start immediately, at least using hybrid technologies. Genuine political will and favorable policies are essential if the renewable energy age is to be fully embraced [12].

In this respect, in Algeria, many studies on wind and solar energy systems have been accomplished [13–33] and several projects achieved:

- Construction of Algeria's first pilot wind power project, a 10 MW facility at Adrar in the southern desert, has been completed in 2011, with the installation of 12 Gamesa 850 kW turbines.

French electrical-engineering company Cegelec built the Adrar plant for the state utility board Sonelgaz. It was originally due to come online in 2012, but the political situation caused various delays.

- Achievement of The Electricity and Gas engineering Company (CEEG) and the German Group Centrotherm/Kinetics signed the contract for the manufacturing of photovoltaic modules of Rouiba éclairage [34].

In this wake, a trade show for the development of national outsourcing in the field of photovoltaic industry was organized. In addition, a call for expression of interest was issued for the construction of an industrial complex specialized in manufacturing silicon cells [34].

On another level, a committee of national integration of renewable energy and a foundation of Mediterranean power grids managers were established [34].

- On 21 December 2011, upon the mandate of SPE, the Electricity and Gas engineering Company (CEEG) conducted the opening and evaluation of financial bids for the achievement of a photovoltaic panel station project in Ghardaia. The contract was awarded to the Group ABB Italy/ABB Algeria who offered the lowest price [34].

Nevertheless, this paper analyzed the environmental and economic benefits of GCRES (grid-connected renewable energy system) used in the arid environment. In Salah and Adrar are taken as two examples of the famous Algerian arid land. The climatic data are used to simulate the long-term implementation of the system.

2. Site description and data characteristics

2.1. Presentation of the studied sites

The two studied sites are Adrar and In Salah where:

- Adrar is a province in southwestern Algeria, named after its capital Adrar. It is the second-largest province, with an area of 424,948 km². It is bordered to the north by the province of Bechar, El Bayadh and Ghardaia; to the west by the province of Tindouf; to the east by the province of Tamanrasset; to the south by Mauritania and Mali [35] (Fig. 1).
- In Salah is an oasis town in central Algeria. It was once an important trade link of the trans-Saharan caravan route. The village is located in the heart of the Sahara Desert region of northern Africa [35] (Fig. 1).

2.2. Meteorological data characteristics

Adrar and In Salah have a hot desert climate (Köppen climate classification BWh), with long, extremely hot summers and short, very warm winters, and averages of just 15 mm (0.59 in) and 16 mm (0.63 in) of rainfall per year, respectively. Summer temperatures are extremely high as they commonly approach 50 °C (122 °F) – with heat indices reaching extreme levels for four months of the year even with the excessively dry air – but temperatures at night fall low enough to be quite bearable, around 30 °C (86 °F). Even in early May or in late September, daytime temperatures can easily rise up to 45 °C (113 °F). Adrar and In Salah experience the same kind of desert heat as Death Valley, California during summertime. Reggane, Adrar and In Salah form the “Triangle of Fire”, as local inhabitants say, due to the extreme desert heat that takes place in the region from May to September.

Thereby, the solar irradiation is among the highest found on the planet, the annual sunshine duration being estimated between 3700 h, and 4000 h. Adrar is often quoted as one of the hottest spots worldwide [35].

In this study, the hourly mean wind speed, global solar irradiance and ambient air temperature values obtained from Meteo-norme 7.1 [36] resource website at two selected sites have been used. The geographical details of the two sites are summarized in Table 1.

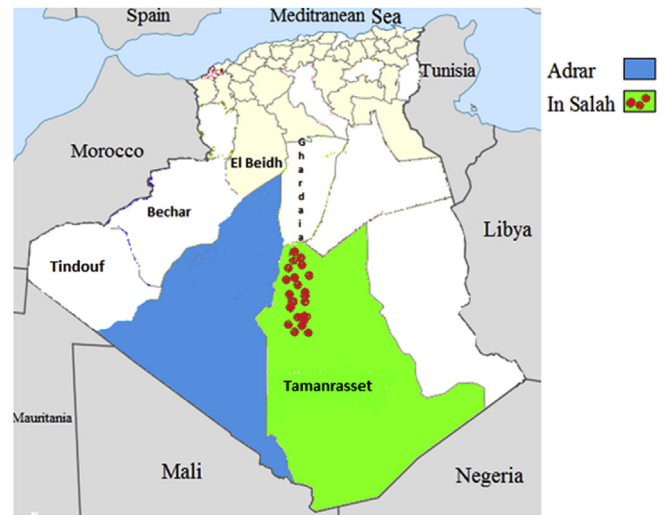


Fig. 1. Map of the two studied sites (Adrar and In Salah).

2.3. Distribution of electricity in the studied sites

In the western south of Algeria, the distribution of electricity is provided by five electricity-operating services Fig. 2:

- Adrar Service operating;
 - Reggane Service operating;
 - Timimoun Service operating;
 - Aoulef Service operating;
 - In Salah Service operating.
- ⚡ The nominal voltages of Service: 30 kV;
⚡ Installed power MV/LV stations: 757,774 kVA;
⚡ Consumption was 221 MVA in 2014.

3. The grid connected renewable energy system configuration

The studied energy system consists of wind and solar energy systems components working together with a non-renewable energy component whereby in this case, the non-renewable energy component is the grid. Fig. 3 shows the configuration of the GCRES in HOMER.

3.1. Load demand

In this work, a simple small village connected to the classical energy distribution network is considered. Nonetheless, it is equipped with a number of devices to the comfort of its inhabitants like hospitals, plant and schools. Additionally, it is assumed to be occupied all over the year and that its devices require the standard rated voltage (220 V AC, 50 Hz).

Fig. 4 shows sample schematic load curves related to the considered community. Typically, in a small village, there are one-peak loads of about 4.6 MW, when people turn on their lights in the evening. The annual energy consumption of the village is

Table 1
Geographical data of the selected sites.

Sites	Latitude (°)	Longitude (°)	Altitude (m)	Climatic classification
Adrar	27.49	−0.11	279	Bwh
In Salah	27.12	2.28	268	Bwh

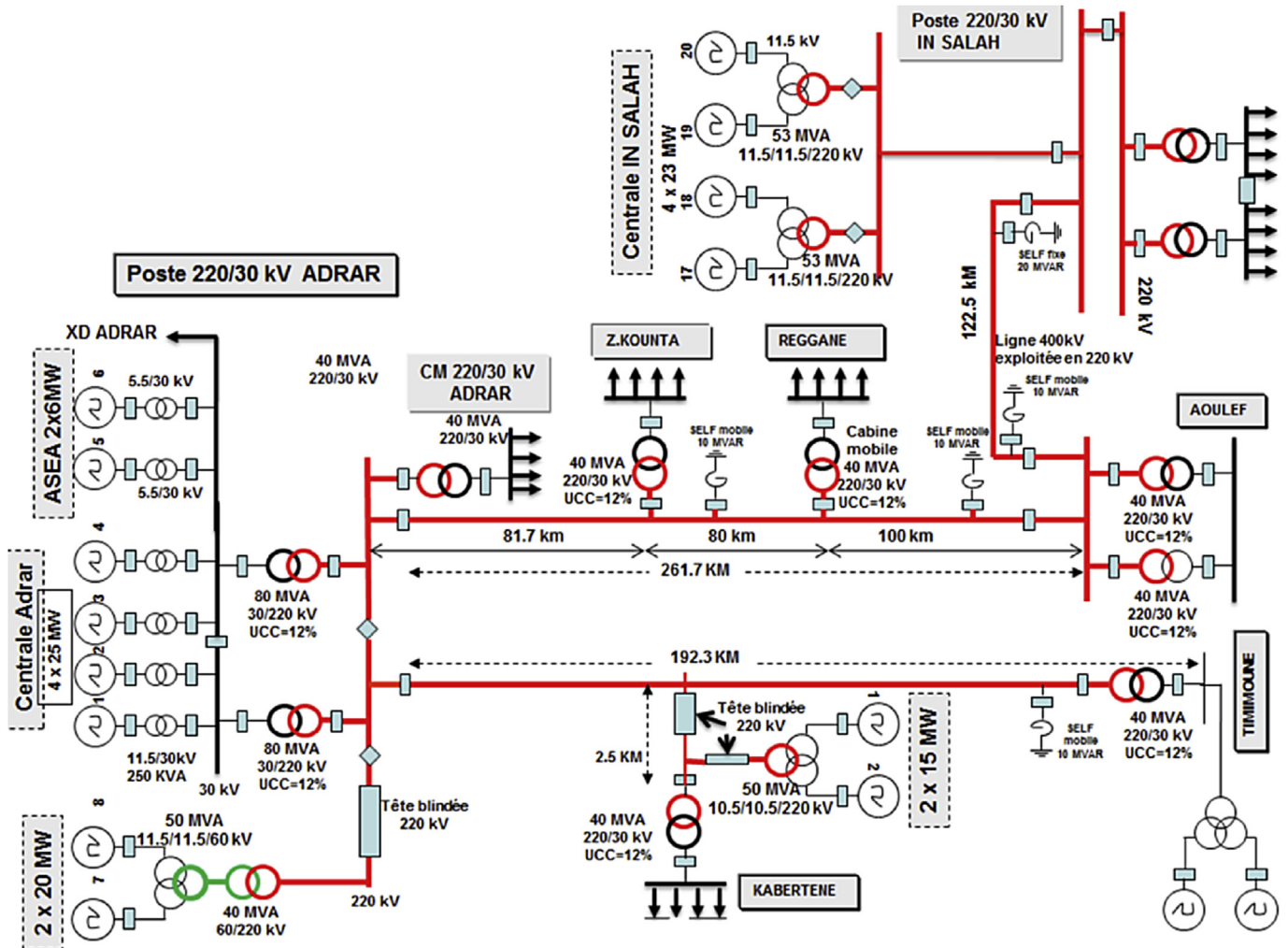


Fig. 2. Adrar-In Salah network scheme.

24496.8 MWh with an average daily energy consumption of 67 MWh.

3.2. Wind energy resources

For the two considered sites, the mean hourly wind speed values obtained from METEONORM 7.1 [36] have been used as input data and the corresponding yearly evolutions are illustrated in Fig. 5(a) for In Salah and in Fig. 5(b) for Adrar. It will be noted that annual wind speed average values of 5.62 m/s and of 7.81 m/s correspond respectively to In Salah and Adrar locations. From the

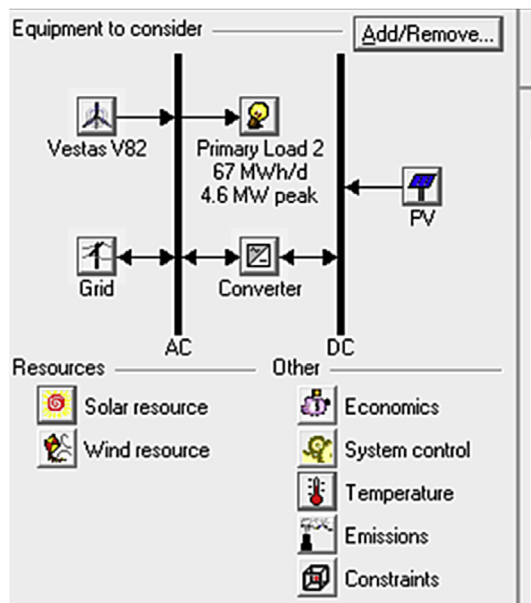


Fig. 3. Configuration of GCRES in HOMER.

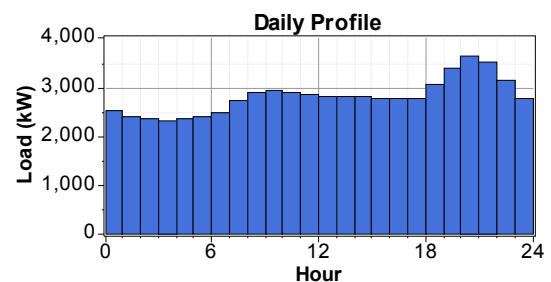


Fig. 4. Hourly load profile.

same figures, it can also be observed that the highest wind velocity occurs during the months of March, April and December. On the other hand, in the other months the wind speed is not low which favors considering the selected sites for wind energy production.

Let us call back that the rotors of the modern wind machines are placed at heights varying between 50 and 100 m, so this data was calculated for an 80 m hub height using boundary layer flow [37]:

$$v_2 = v \cdot \frac{\log\left(\frac{H}{z_0}\right)}{\log\left(\frac{z}{z_0}\right)} \quad (1)$$

Where

H is the hub height of the wind turbine generator (m)
 z is the anemometer height (m)
 v is the wind speed at the anemometer height (m/s)
 z_0 is the surface roughness lengths defined in Ref. [37].

At 80 m height, the average wind speed becomes 6 m/s while at 10 m it is only 5.010 m/s (see Fig. 6).

Fig. 7 shows that the wind speed is mainly distributed between 4 m/s and 8 m/s for In Salah and between 3 m/s and 8 m/s for Adrar. From the same figure, it is demonstrated that wind speed data fit a Weibull distribution with a scale parameter $k = 2.01$ and a shape factor $c = 8.82$ m/s for In Salah and $k = 2.03$ and a shape parameter $c = 6.34$ m/s for Adrar. As seen from Fig. 8, wind shortage is experienced during March and April by the two studied sites.

3.3. Solar energy resources

For both considered site, the hourly mean global solar radiation data has been obtained from METEONORM 7.1 [36].

The data shows that the exposure duration is longer than 4516 h per year. Scaling was done on this data to consider the long-term average annual resource (5.72 and 6.12 kWh/m² per day) respectively for Adrar and for In Salah. HOMER introduces the clearness

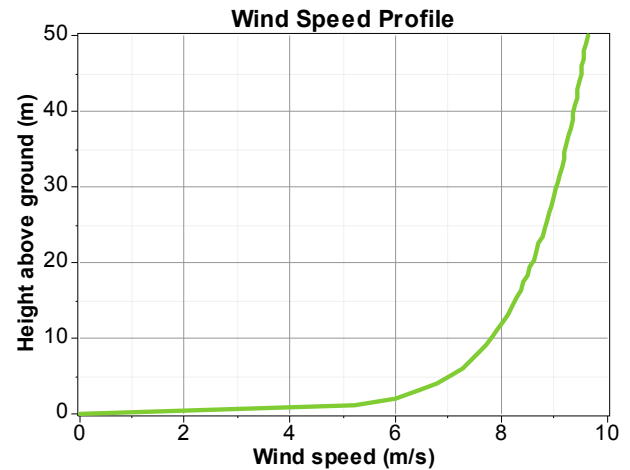


Fig. 6. Wind speed variation with Height.

index from the location (latitude and longitude) information of the site under investigation. In Fig. 9(a) are presented the yearly distribution of the daily solar irradiation in kWh/m² per day and the clearness index curve over the period of the whole year. The monthly mean solar irradiation value varies from 0.70 to 3.5 MJ/m²/day for Adrar and from 0.8 to 3.8 MJ/m²/day for In Salah. The maximum value was seen in July while the minimum one was seen in December. As seen from the Fig. 9(a), Adrar and In Salah regions has more solar resource during the summer season than during the winter. Considering the annually variations, the system sensitivity analysis is done by the consideration of two values around the mean, which are 4.5, 5.5, 5.72, 7, 8 kW/m²/day for Adrar and 3, 4, 5, 6.12, 7, 8 kW/m²/day for In Salah. Therefore, Fig. 9(b) shows the hourly global radiation data and Fig. 9(c) shows the Hourly ambient air temperature data evolution over the year for the two considered sites. From Fig. 9(c) appears that the annual average temperatures are 26 °C and 24.3 °C respectively for In Salah and Adrar. It is observed also that the maximum value was seen in summer and the minimum one was seen in winter.

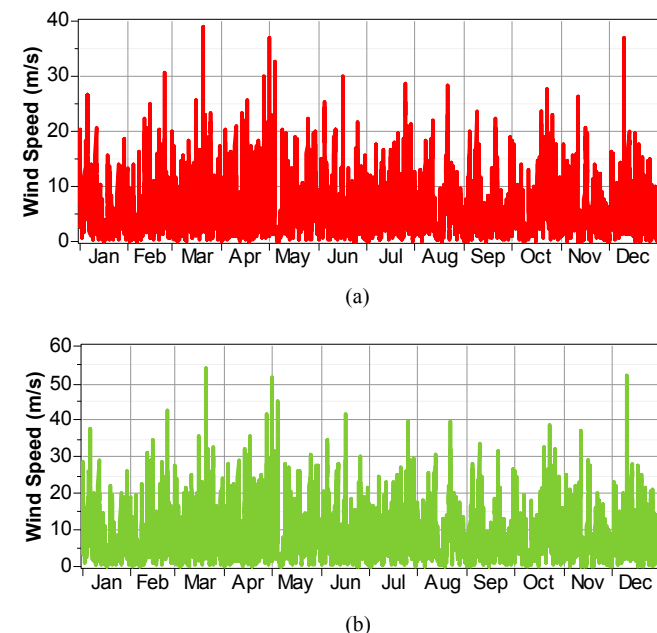


Fig. 5. (a). Hourly wind speed in In Salah. (b) Hourly wind speed data in Adrar.

3.4. Grid connected renewable energy system

The GCRES consists mainly of grid, PV (photovoltaic) array, an inverter and wind turbines. Otherwise, for the economic analysis, HOMER takes into account the initial cost, replacement cost, and operating and maintenance cost.

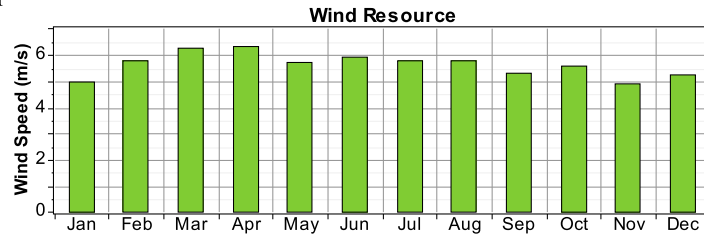
• Wind energy conversion system

Availability of electrical energy from the wind turbine depends mainly from the available energy potential. Hence, in this work, Vestas V82 wind turbine model is considered (Fig. 10). It has a rated capacity of 1.65 kW and provides 690 V AC as output with 50 Hz frequency. Cost of one unit is considered to be 1,262,190\$ while replacement and maintenance costs are taken as 1,262,190\$ and 2000\$. To allow the simulation program find an optimum solution, provision for using 0 (without turbine), 1 or 2 units is given. Lifetime of the wind turbine is taken to be 20 years.

• Photovoltaic energy conversion system

The ALPV80-125 M-36 modules made in Algeria by the ALPV Company [38] constitute the photovoltaic arrays used in the present study. The lifetime of the photovoltaic modules are estimated

a. In Salah



b. Adrar

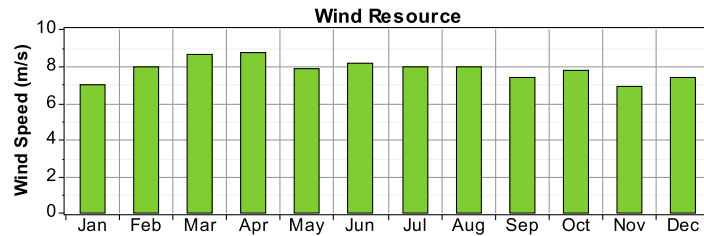
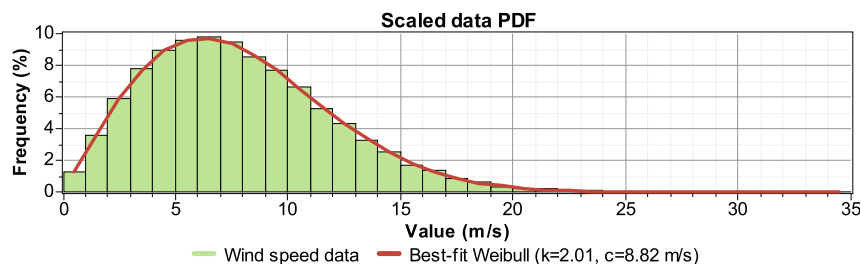


Fig. 7. Average monthly wind speed data.

a. In Salah



b. Adrar

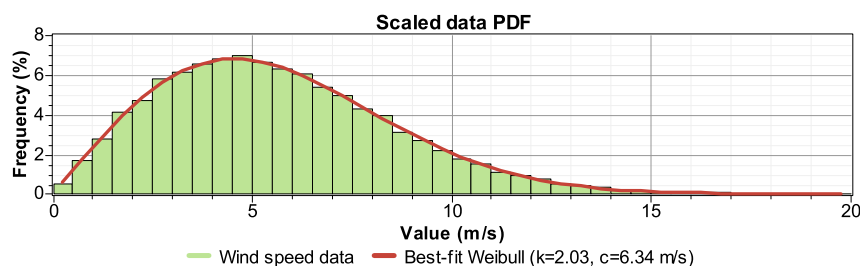


Fig. 8. Wind speed probability distribution function.

equal to 25 years, a correction factor equal to 0.9 is considered to model 10% of losses due to dust, module aging, loss of cables and all difference between the simulated and real conditions. The initial cost of photovoltaic arrays is \$4.6 per watt [16] considering a more optimistic system, the cost of installation, replacement and maintenance of a 1 kW solar energy is evaluated equal to \$4600. In this study, the photovoltaic array size is varied in term of power equal to zero, 1500 kW and to 2000 kW.

- A power converter

A converter is required for systems in which DC components serve an AC load or vice versa. The HOMER software considers a converter as an inverter (DC to AC), rectifier (AC to DC) or both. For 1 kW system the installation and replacement, costs are taken as

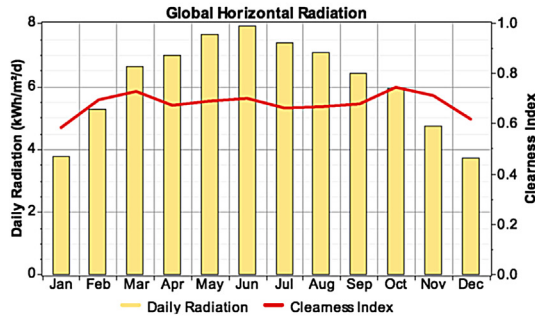
\$800 and \$750 respectively. Three different sizes of converter (0, 1500, 2000 kW) are considered as input in the model. Lifetime of a unit is considered to 25 years with an efficiency of 90%.

- Grid

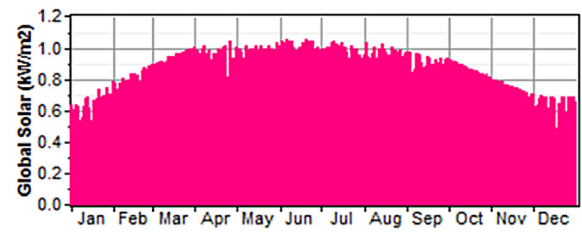
In the present case, the grid is a principal source by which it acts with the two renewables energy system to satisfy continuously the load demand. Therefore, there is grid power price, which is the electricity, bought from the grid and the price of electricity sold to the grid, also known as FIT (feed in tariff). The electricity tariff for the wind and solar system in Algeria is 300% of base rate.

Table 2 illustrated the samples of FIT application in Algeria [24]. It can be remembered that Algeria is depending on base rate and technology.

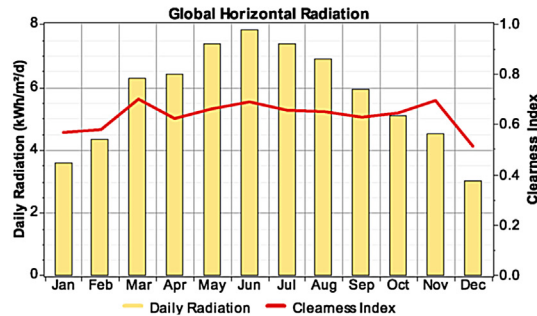
1- In Salah



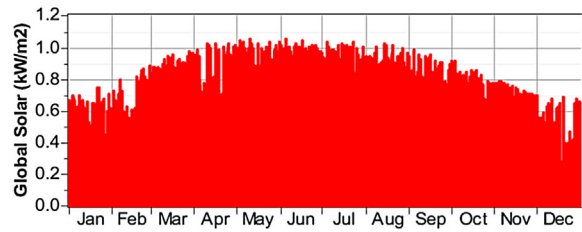
1-In Salah



2- Adrar



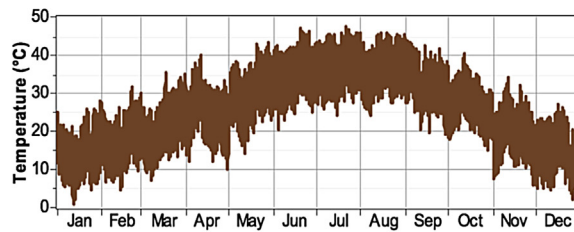
2- Adrar



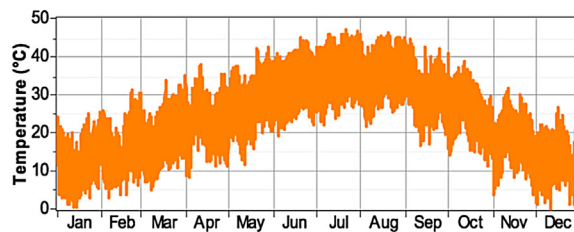
(a)

(b)

1-In Salah



2- Adrar



(c)

Fig. 9. (a) Average daily global solar radiation data. (b) Hourly global radiation. (c) Hourly ambient temperature data.

4. Assessment criteria

Environmental and economic aspects are the two of important aspects of sustainability. To evaluate the sustainable performance of the GCRES, these two impacts should be considered. In this section, emission and renewable fraction are discussed as environmental indicators, and cost and the payback period for economic indicators.

However, HOMER assesses the technical viability of the wind and solar energy system by determining when the system participates to serve the load adequately. Then, it estimates the NPC of the

system and classes all the system according to total net present cost NPC [24].

4.1. Emission

Emission of a GCRES includes carbon dioxide, sulfur dioxide and nitrogen oxides. The emission is used as viable headline assessment for the environmental domain. The calculated result of emission can be reliable and feasible, due to the solar radiation and wind speed data used in the simulation are reliable. The estimating method is based on the User Manual of the software HOMER [39].

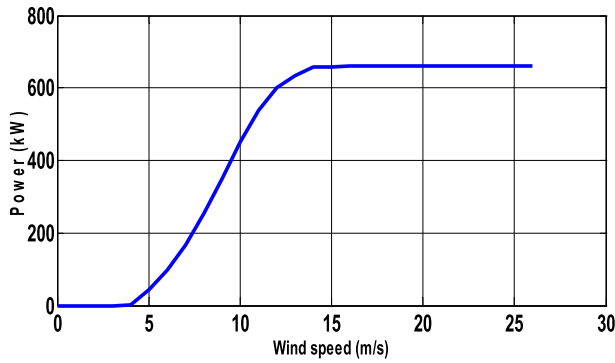


Fig. 10. Wind turbine Vestas V80 versus wind speed.

Table 2

Samples of FIT application in Algeria.

	% of base rate	\$/kWh
Base Rate		4.82
Wind Energy	300%	14.5
Solar Photovoltaic	300%	14.5
Concentrating Solar	200%	9.57
Hydro	100%	4.82
Waste	200%	9.57

4.2. Renewable energy fraction

Renewable fraction has representativeness for environmental assessment. It can be in turn divided into components fraction such as PV fraction (f_{pv}) and wind fraction (f_{WG}). In the study of Celik [40], the two fractions are calculated by the following equations,

$$f_{PV} = \frac{E_{PV}}{E_{TOT}} \quad (2)$$

$$f_{WG} = \frac{E_{WG}}{E_{TOT}} \quad (3)$$

where E_{pv} , E_{WG} and E_{TOT} are respectively the energy generation by photovoltaic, energy generation by wind generator and total energy generation.

4.3. NPC (net present cost)

The total NPC assesses all costs that meet with the lifetime. The costs include initial costs, replacement cost, operating and maintenance costs and cost of electricity from the grid. Total NPC is calculated by using the following equation:

$$NPC = \frac{C_{ann,tot}}{CRF(i, R_{proj})} \quad (4)$$

Where $C_{ann,tot}$ is the total annualized cost [\$/yr];

CRF is the capital recovery factor;

i is the interest rate [%];

R_{proj} is the project lifetime [yrs].

The nominal interest rate and annual inflation rate in Algeria as of 2014 are 5.9% and 1.8% respectively [24]. Therefore, the annual real interest rate is 2.25%.

Table 3

Categorized optimization Results.

A) IN SALAH

Calculate

Simulations: 8 of 8

Progress:

Sensitivities: 400 of 400

Status: Completed in 30 seconds.

Sensitivity Results

Optimization Results

Sensitivity variables

Global Solar (kWh/m²/d)

6.12

Wind Speed (m/s)

5.62

Rate 1 Power Price (\$/kWh)

0.7

Rate 1 Sellback Rate (\$/kWh)

0.4

Double click on a system below for simulation results.

Categorized

Overall

Export...

Details...

	PV (kW)	V82	Conv. (kW)	Grid (kW)	Initial Capital	Operating Cost (\$/yr)	Total NPC	COE (\$/kWh)	Ren. Frac.
	1500	1	1500	5000...	\$ 8,655,690	12,859,353	\$ 173,041,3...	0.553	0.31
			1	5000...	\$ 1,262,190	13,829,908	\$ 178,054,8...	0.563	0.19
	1500		1500	5000...	\$ 7,393,500	16,173,276	\$ 214,142,2...	0.684	0.12
				5000...	\$ 0	17,148,460	\$ 219,214,8...	0.700	0.00

B) Adrar

Sensitivity Results

Optimization Results

Sensitivity variables

Global Solar (kWh/m²/d)

5.72

Wind Speed (m/s)

7.81

Rate 1 Power Price (\$/kWh)

0.7

Rate 1 Sellback Rate (\$/kWh)

0.4

Double click on a system below for simulation results.

☒ Categorized☐ Overall

Export...

Details...

	PV (kW)	V82	Conv. (kW)	Grid (kW)	Initial Capital	Operating Cost (\$/yr)	Total NPC	COE (\$/kWh)	Ren. Frac.
	1500	1	1500	5000...	\$ 8,655,690	11,939,012	\$ 161,276,3...	0.515	0.36
			1	5000...	\$ 1,262,190	12,784,977	\$ 164,697,1...	0.526	0.26
	1500		1500	5000...	\$ 7,393,500	16,297,487	\$ 215,730,0...	0.689	0.11
				5000...	\$ 0	17,148,460	\$ 219,214,8...	0.700	0.00

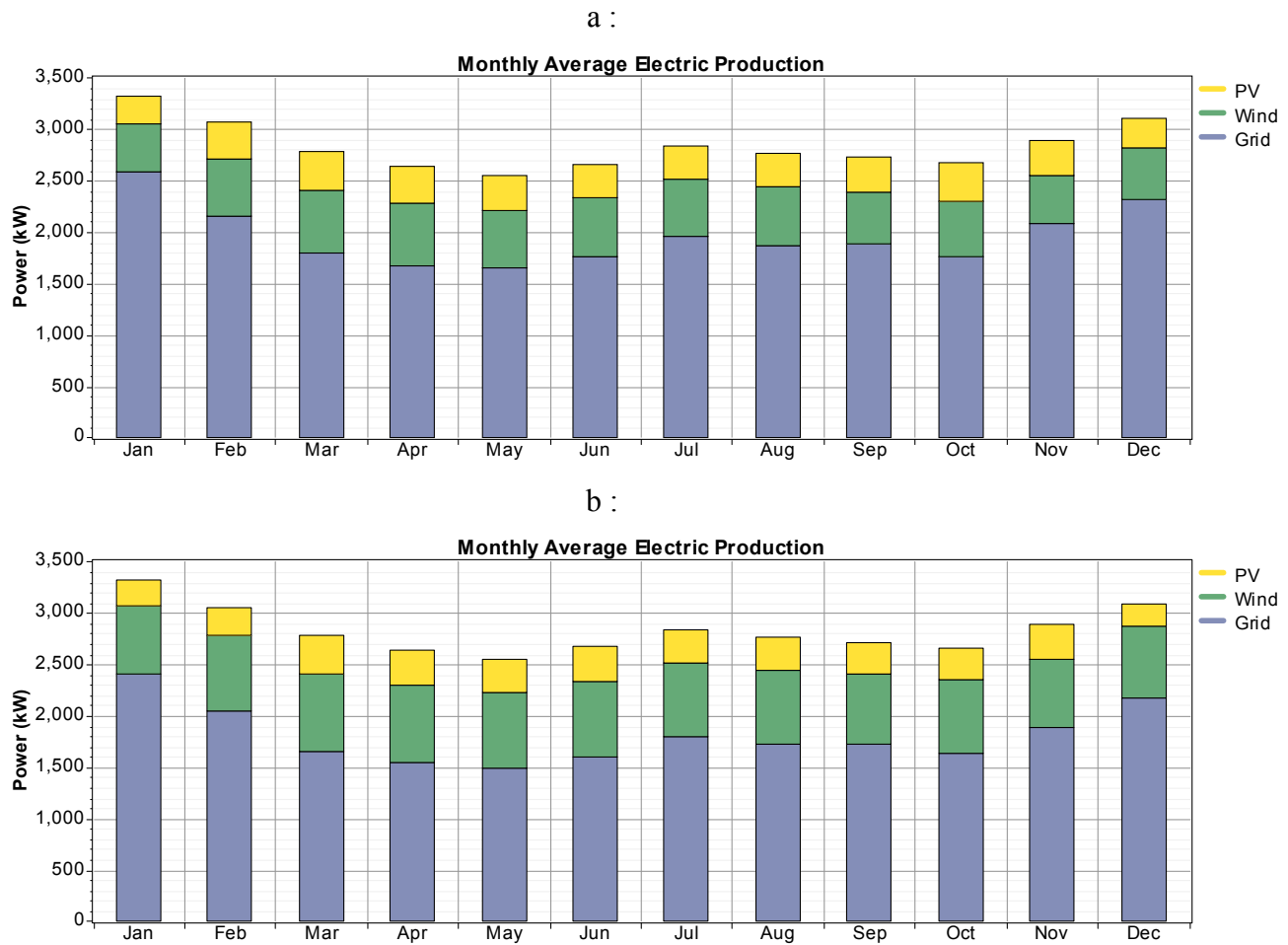


Fig. 11. Average Monthly Electric Production from each source (a: In Salah, b: Adrar).

4.4. Payback period

The payback period is an indication of how long it would take to recover the difference in investments costs between the current system and the base case system. In this case, the current system would be the GCRES and the base case system would be the grid-only system.

5. Results and analysis

5.1. Optimisation results

The categorized result for In Salah and Adrar are presented respectively (Table 3a and b), in which the two systems (the grid

only and the GCRES) architecture types are found, is listed in increasing order of total NPC (Net Present Cost). There are two different feasible systems architectures starting from GCRES to the grid only.

From results presented in Table 3a and b it is observed that for both site the GCRES with 1 unit of 1.65 kW wind turbine and 1500 kWc PV capacity is the most viable system to be applied as it has the lowest NPC.

Otherwise, in Fig. 11a and Fig. 11b are presented respectively the electricity amount produced by the GCRES if installed in In Salah and in Adrar site.

From the results presented in the two figures it can be seen that the GCRES Implemented under the electricity load of

Table 4
Economic comparison between GCRES and grid only system.

	PV (kW)	V82	Grid (MW)	Initial capital	Total NPC	
Base Case			5000	\$0	\$219,214,880	
Current System	1500	1	5000	\$1,262,190	In Salah \$173,041,376	Adrar \$161,276,336
Metric	Value					
	In salah		Adrar			
Present worth (\$)	46,173,528		57,938,580			
Annual Worth (\$/yr)	3,612,004		4,532,345			
Return on investment (%)	49.6		60.2			
Internal rate of return (%)	51.1		61.7			
Simple Payback (yrs)	1.98		1.62			
Discounted payback (yrs)	2.15		1.76			

Table 5

Yearly emissions of GCRES and Grid only system.

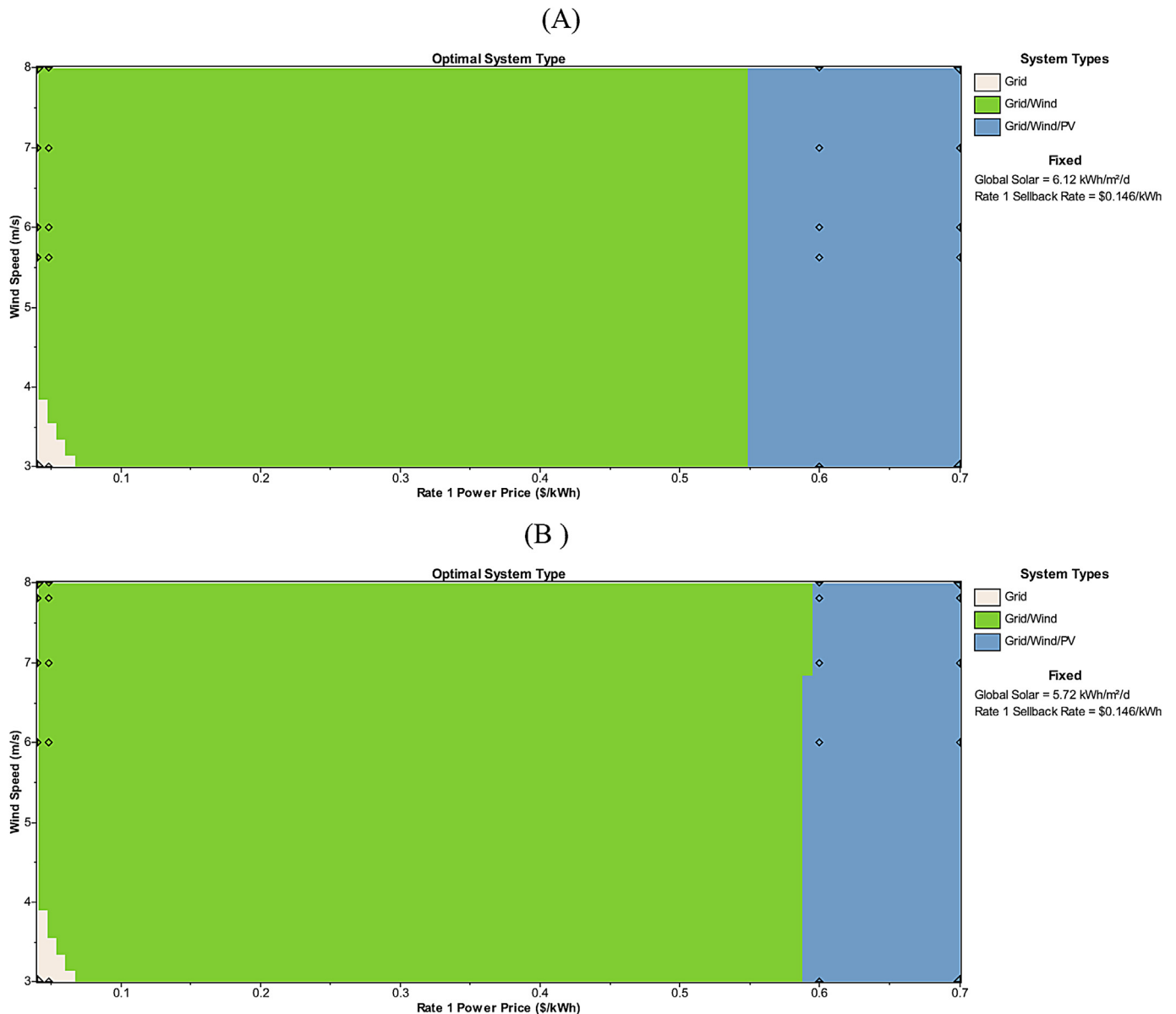
Site	In salah	Adrar	In salah & Adrar
Pollutant	Emissions of the GCRES (kg/yr)	Emissions of the GCRES (kg/yr)	Emissions of the grid only system (kg/yr)
Carbon dioxide	10,818,335	9,987,056	15,482,610
Sulfur dioxide	46,902	43,298	67,124
Nitrogen oxides	22,938	21,175	32,827

67 MWh/day, the PV array produces: 2907 MWh/year (fpv = 12%) for In Salah and 2701 MWh/year (fpv = 11%) for Adrar by cons, the wind system produces respectively almost (5190 MWh/year) and (6215 MWh/year) for In Salah and Adrar. The wind systems energy production constitutes about 21% of the system total energy production (24,785 MWh/year) for In Salah and about 25% of that produced in ADRAR (24,802 MWh/year). In another word, for this system, the grid purchases share of 67% (15,704 MWh/year) for In

Salah and 64% (15,818 MWh/year) for Adrar, of the total energy production.

5.2. Grid –only system

The NPC of a grid only system is 15,031,877 \$ under consideration of cost of 0.048 \$/kWh. The total electricity produced by the grid and used to cover the electric load is 24,497 MWh/yr.

**Fig. 12.** Result of sensitivity analysis (A: In Salah, B: Adrar).

5.3. GCRES system

At FIT rate equivalent to PV–Wind-grid system of 0.145 \$/kWh, the GCRES which consists of one unit of 1.65 kW wind turbine and 1500 kWc PV capacity is the most viable system to be implemented in both sites under the previous conditions consideration. Hence, the NPC is evaluated equal to \$173,041,376 for In Salah and which is about 12% lower than the NPC of Grid only system consideration. Results presented in Fig. 11a shows that the electricity produced by the renewable energy system and by the grid are respectively about 31% and 69%.

Otherwise, the NPC relative to the Adrar site is \$161,276,336, which is approximately 16% lower than the NPC of the Grid only system. Fig. 11b shows that the electricity produced by the renewable energy system and by the grid are respectively about 36% and 64%.

These show that the renewable fraction of the system is small compared to the conventional grid which means the avoid of

problems that can result by the connection of renewable energy system to the electrical grid. In the case of no wind and no solar, the electric grid only will ensure energy production to supply the electrical load need. Therefore, the GCRES is feasible from economic point of view. In addition, payback period for present system is obtained as 1.98 years for In Salah and 1.62 years for Adrar (Table 4) which ensures approximate 2 years of net income for a project lifetime of 25 years.

Furthermore, the amount of yearly emissions of the GCRES and the grid only system are given in Table 5 for In Salah and Adrar. In this study, the amount of emission of CO₂, SO₂ and that of nitrogen oxides are reduced to 10,818,335 kg/year, 46,902 kg/year, and 22,938 kg/year respectively for In Salah. Therefore, for Adrar they are reduced to 9,987,056 kg/year, 67,124 kg/year, and 32,827 kg/year respectively. These values are very much lower than compared to those emitted by the grid system. Hence, this confirm that the GCRES considered in this study is beneficial in the context of clear environment, too.

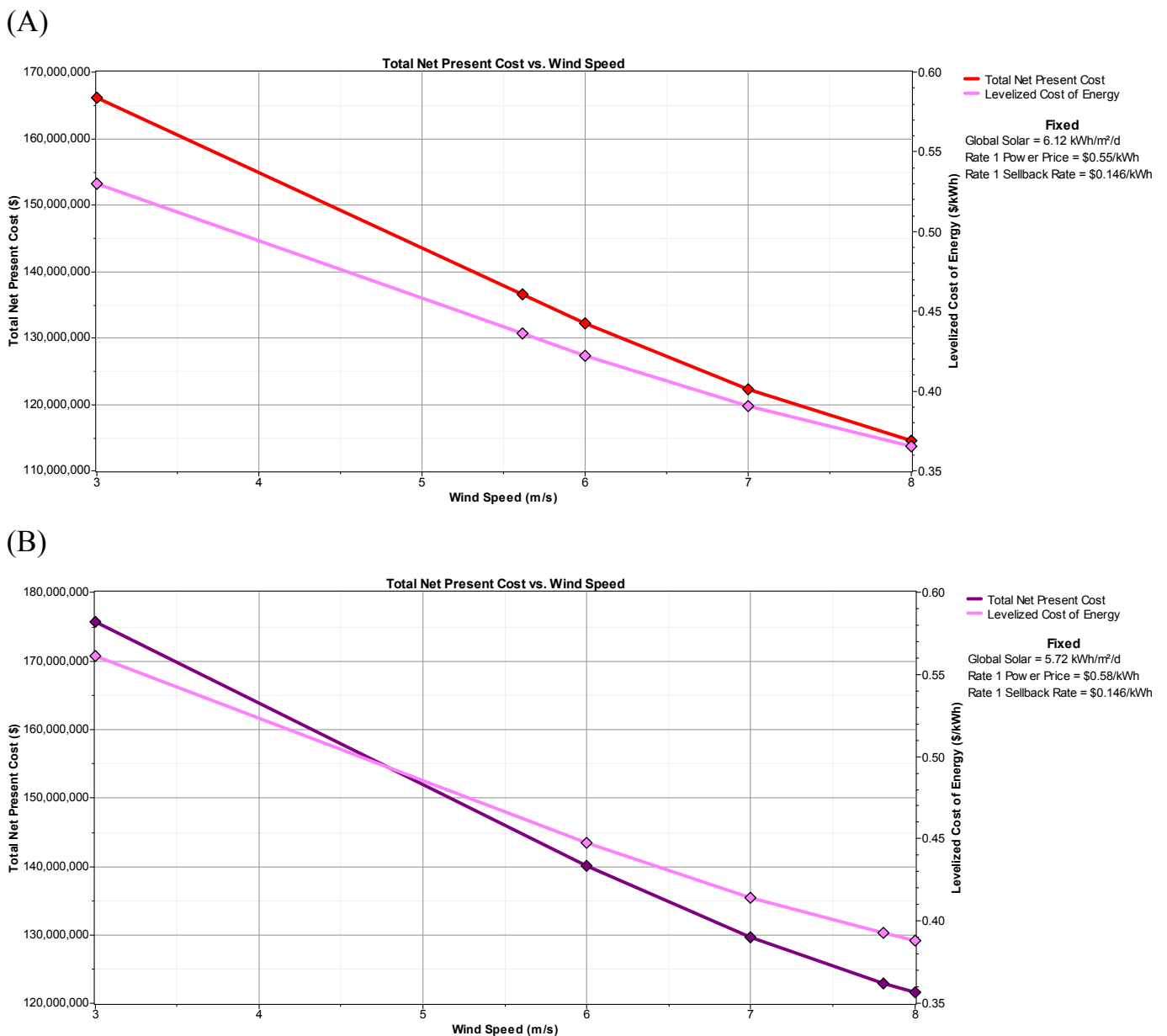


Fig. 13. The relationship between cost and wind speed (A: In Salah, B: Adrar).

6. Sensitivity analysis

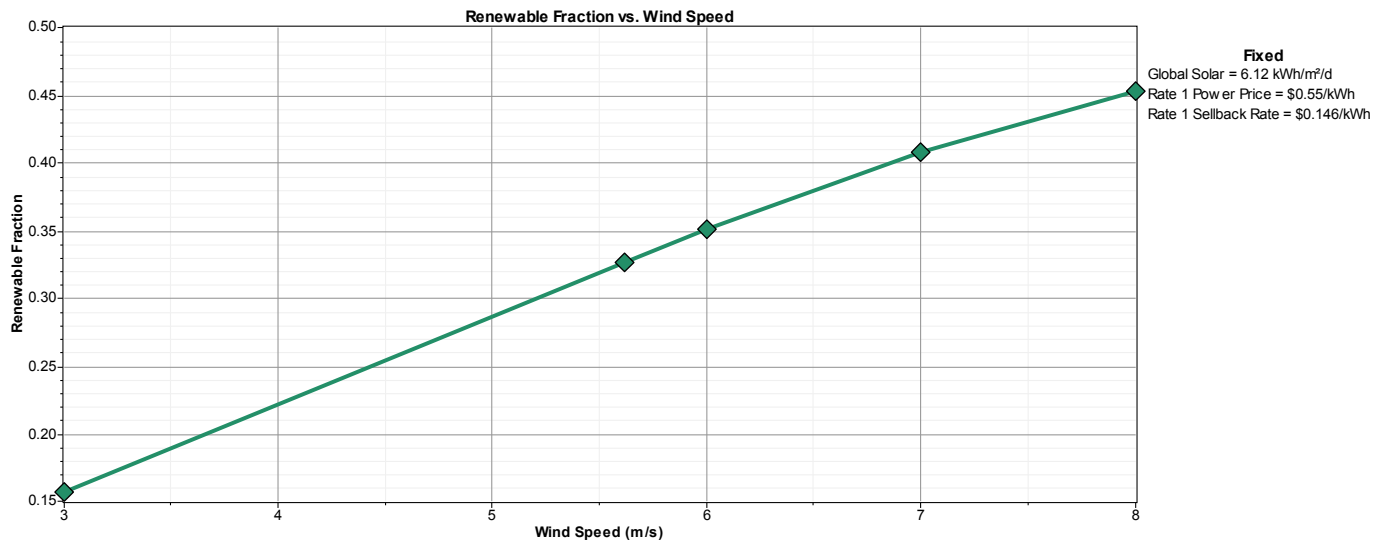
Sensitivity results are analyzed to study the impacts of variation in the solar irradiation and wind speed on the performance of the GCRES. The simulation software simulates the long-term implementation of the GCRES based on their respective search size for the predefined sensitivity values of the components. The emissions, renewable fraction, NPC and COE (cost of energy) are simulated based on the four sensitivity variables: wind speed (m/s), solar irradiation ($\text{kWh}/\text{m}^2/\text{day}$), grid electricity price ($\$/\text{kWh}$) and the sellback rate ($\$/\text{kWh}$). For all of the sensitivity values HOMER simulates all the systems in their respective predefined search space.

In this study, the sensitivity variables of solar irradiation are defined as: $G = 3, 4.5, 6.12, 7, 8 \text{ kWh}/\text{m}^2/\text{day}$ for In Salah and

$G = 4.5, 5.5, 5.72, 7, 8 \text{ kWh}/\text{m}^2/\text{day}$ for Adrar, while wind speed are: $v = 3, 5.62, 6, 7, 8 \text{ m/s}$ for In Salah and $v = 3, 6, 7, 7.81, 8 \text{ m/s}$. Moreover, the grid electricity price (P) and sellback rate (S) are also defined as a sensitivity variable ($P = 0.04, 0.4, 0.5, 0.6, 0.7 \text{ \$/kWh}$) and ($S = 0.146, 0.2, 0.3, 0.4, 0.5 \text{ \$/kWh}$) for Adrar and In Salah. A total of 625 sensitivity cases were tried for each system configuration. The simulation time was 52 s on personal computer with Intel CORE i3 CPU of 3.33 GHz and 3 GB RAM.

The feasibility of the system is analyzed through the sensitivity results in terms of solar irradiation, wind speed, grid electricity price and sellback rate. Here, the feasibility of hybrid power system is analyzed based on emission reduction and cost saving. This type of sensitivity analysis of the systems provides information that a particular system would be optimal at certain sensitivity variables [41].

(A)



(B)

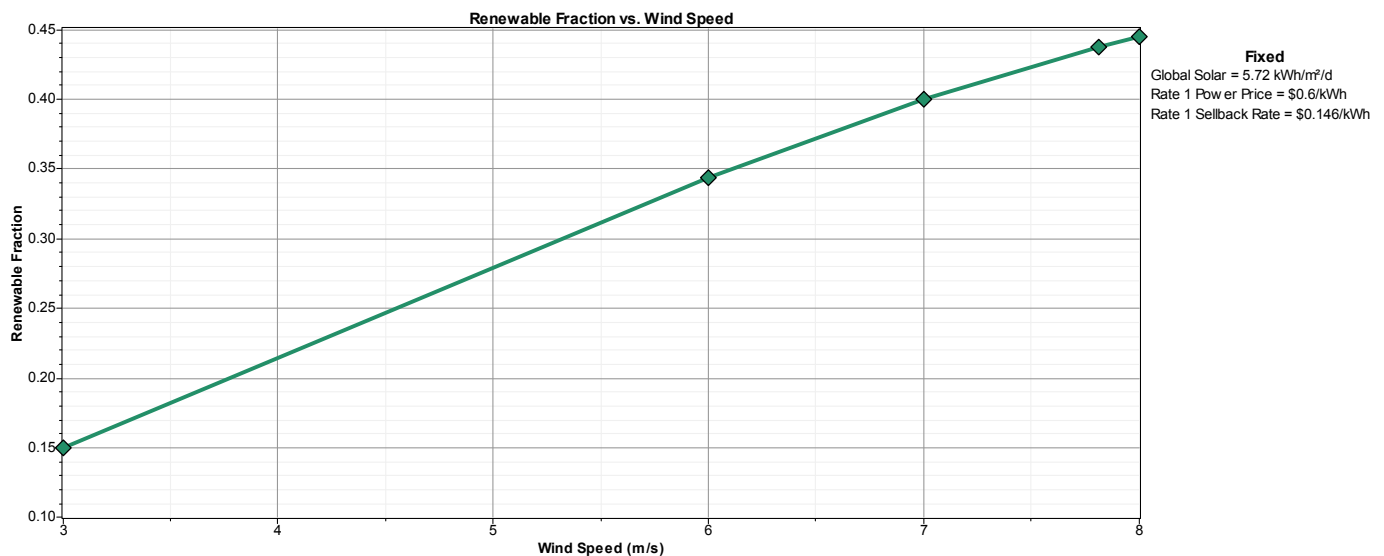


Fig. 14. The relationship between RF and wind speed (A: In Salah, B: Adrar).

The system is feasible when the grid electricity price is more than 0.55 \$/kWh and 0.58 \$/kWh respectively for In Salah and Adrar. Under these conditions, the RF can be between 0.15 and 0.45. A GCRES is feasible when global solar irradiation is more than 6.12 kWh/m²/day for In Salah and 5.72 kWh/m²/day for Adrar. The wind speed is between 3.0 and 8 m/s, the sellback rate is more than 0.0146 and the grid electricity price is more expensive than 0.55 \$/kWh and 0.58 \$/kWh respectively for In Salah and Adrar (Fig. 12). Based on the optimization results, wind energy production shows a bigger proportion of energy generation than solar. While the solar power occupies less than 12% and 11% of the total energy generation respectively for In Salah and Adrar, wind power occupies approximately half. Therefore, the wind energy resource has more impacts on the implementation. Figs. 13–16 reflect the cost, renewable fraction and emissions variation dependent on the sensitivity variable wind speed. The NPC and COE of the GCRES

reduces when the wind speed increases from 3.0 m/s to 8.0 m/s (Fig. 13). Simultaneously, as seen in Fig. 14, renewable fraction rise sharply from 0.15 to 0.45 when wind speed increases from 3.0 m/s to 8 m/s. As shown in Figs. 15 and 16, the main emissions of carbon dioxide, sulfur dioxide and nitrogen oxide decrease sharply when the wind speed decreases.

7. Conclusion

In this paper, grid-connected renewable energy system (GCRES) is considered to be installed in two Algerian sites characterized by an arid and hot climate according to the Koppen–Geiger climate classification [42] that will be valid for all sites characterized by the same climatic characteristics. By using the HOMER software tools, an economic and environmental analysis of two different configurations that are the grid only system and the wind–PV-grid

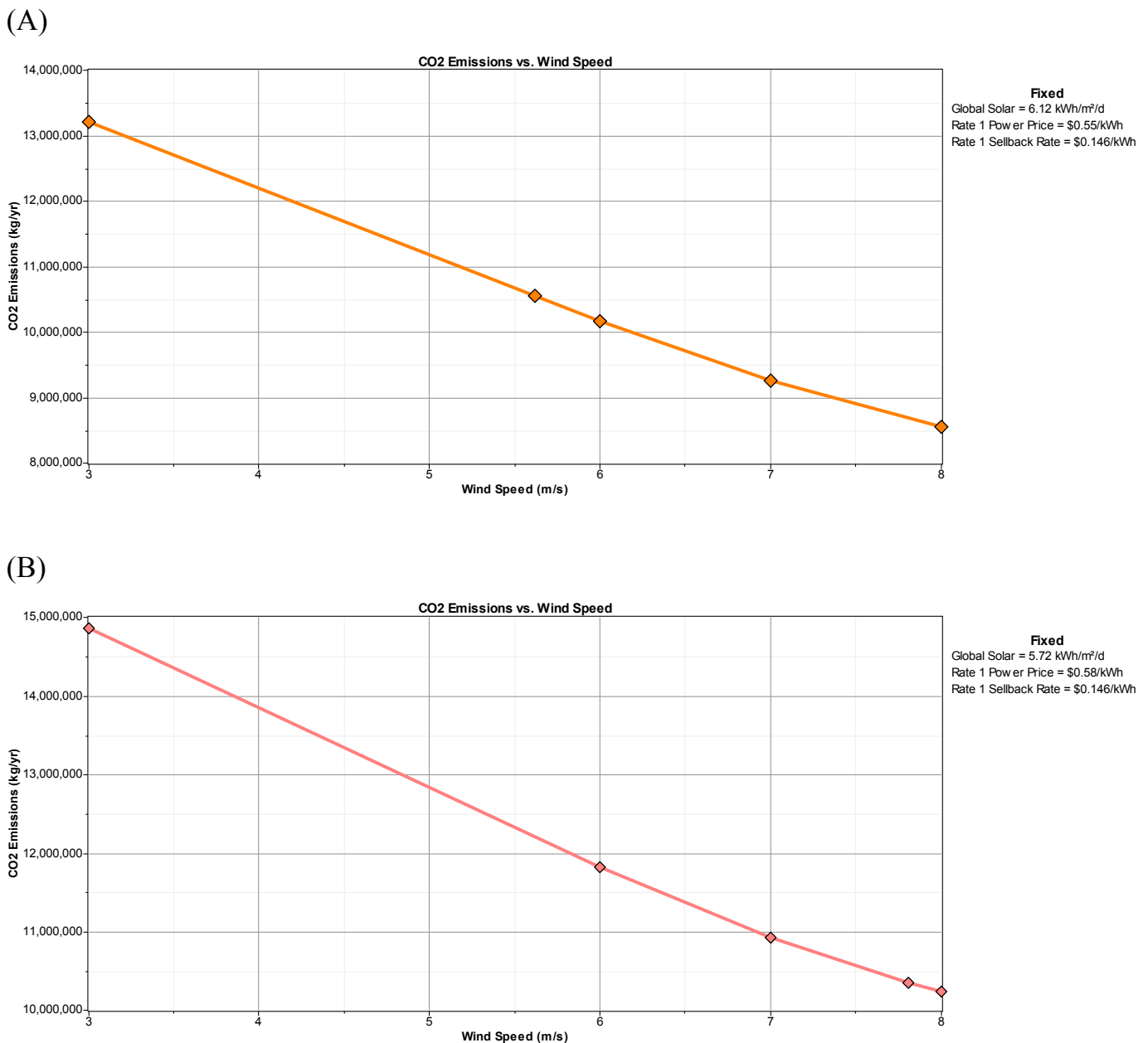
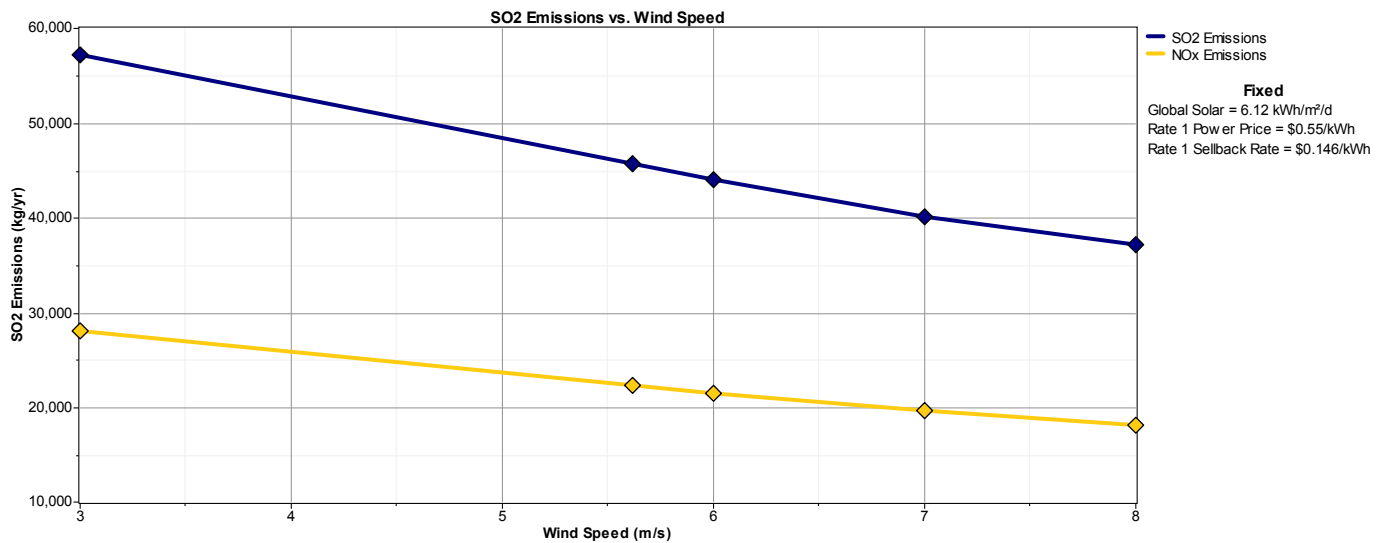


Fig. 15. The relationship between CO₂ emissions and wind speed (A: In Salah, B: Adrar).

(A)



(B)

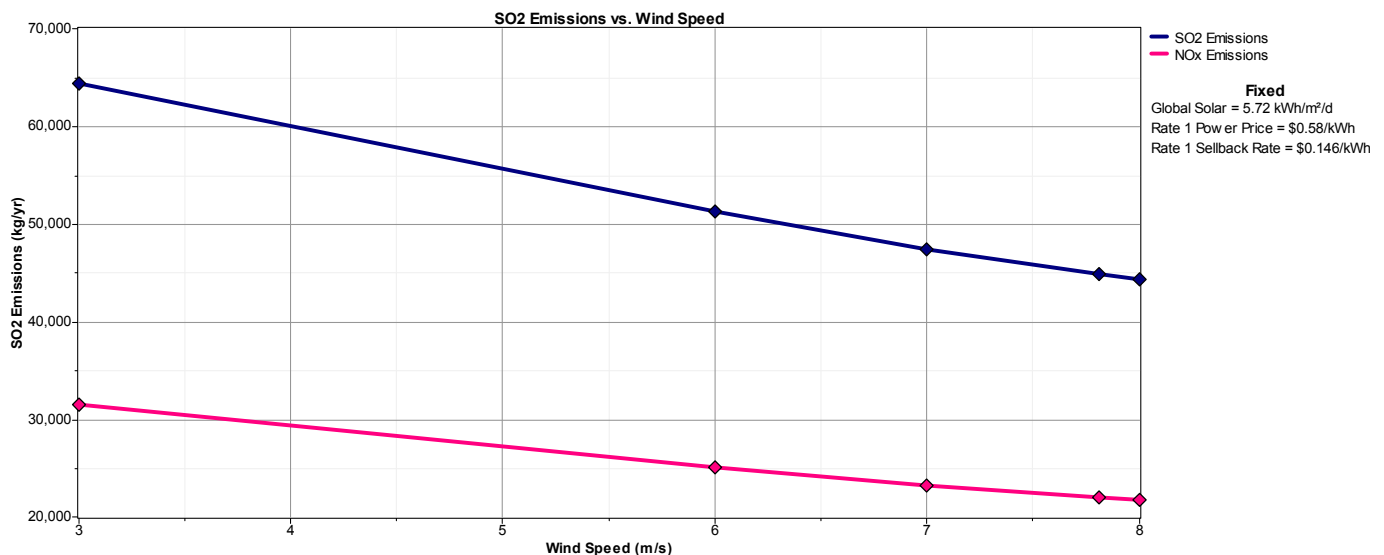


Fig. 16. The relationship between sulfur dioxides and nitrogen oxide emissions, and wind speed (A: In Salah, B: Adrar).

system has been presented. A large-scale load data and the mean hourly wind speed and solar radiation data were used for the assessment. The assessment criteria chosen are emission and renewable fraction as environmental indicators, and the cost and the payback period as economic indicators. Thus, the simulation showed that the GCRES is the most viable system economically. At fixed FIT rate of 0.145\$, the NPC of grid connected one unit of 1.65 kW wind turbine and 1500 kW of is \$173,041,376 for In Salah and is \$161,276,336 for Adrar with the RF of the optimized system (GCRES) of 31%, and 36% respectively for In Salah and Adrar. However, for the same FIT rate cited previously, the payback period for GCRES is approximately 2 years for both considered sites. From the environmental standpoint, it is observed that the optimum results of GCRES uses show that a 31% and 36% respectively for In Salah and Adrar emissions reduction including CO₂, SO₂ and NO_x is realized. In addition, through a set of sensitivity analysis it is found that the wind speed value has more effects on

the environmental and economic performance of a GCRES which agree with the result obtained by Shamim Kaiser M et al. [43] and Ghiani, Emilio [44].

To sum up, according to the data used in the present study, the renewable energy resource (including solar and wind energy) in In Salah and in Adrar is abundant in some period but relatively less in the other. It can be forecasted that the hybrid renewable energy system needs more grid electricity as complementarities.

Finally, the overall system produced kWh price would be low if the turbine, the PV modules and battery cost decreases, which is confirmed by Kalinci, Yildiz et al. [45].

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